

Acoustic similarities, articulatory uniqueness: Speech production mechanisms in individuals with congenital lip paralysis

Anne Hermes¹, Ivana Didirková², Philipp Buech¹, Gilles Vannuscorps³

¹Laboratoire de Phonétique et Phonologie, UMR 7018, Université Sorbonne Nouvelle, CNRS, Paris, France

²Praxiling, UMR 5267, Université de Montpellier Paul-Valéry, CNRS, Montpellier, France

³Psychological Sciences Research Institute/IONS, Université catholique de Louvain, Belgium
anne.hermes@sorbonne-nouvelle.fr, ivana.didirkova@univ-montp3.fr,
philipp.buech@sorbonne-nouvelle.fr, gilles.vannuscorps@uclouvain.be

Abstract

Moebius syndrome is a rare congenital neuromuscular disorder characterized by facial paralysis, which can severely impact speech. Despite these constraints, some individuals with congenital lip paralysis develop very natural speech intelligibility by employing strategies that remain underexplored. This study examines the acoustic-articulatory adaptations in three French-speaking individuals with Moebius syndrome compared to three controls. Using articulatory and acoustic data, we analyzed speech patterns in disyllabic words. Results indicate that Moebius speakers achieve acoustically similar vowel targets to controls despite significant articulatory differences, including increased tongue movement to compensate for lip paralysis. These findings highlight the adaptability of the speech motor system, supporting the notion of motor equivalence in achieving intelligible speech.

Index Terms: congenital lip paralysis, Moebius syndrome, acoustics-articulation, adaptations, motor equivalence, speech therapy

1. Introduction

1.1. Moebius syndrome

1.1.1. General

Moebius syndrome is a rare congenital neuromuscular disorder affecting approximately 0.00002 to 0.002% of the world's population. This condition is characterized by the absence or underdevelopment of the 6th and 7th cranial nerves, which are responsible for controlling horizontal eye movements and facial muscles [1]. One of the most prominent symptoms of Moebius syndrome is bilateral facial paralysis. This paralysis significantly impacts facial motor skills, affecting essential functions such as expressions and eating. Fig. 1 is showing an individual with Moebius syndrome for different tasks involving the lips.

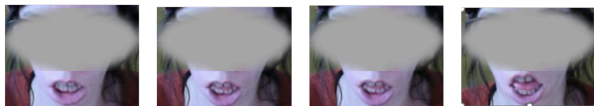


Figure 1: Individual with Moebius syndrome trying to (A) close the lips as much as possible, (B) stretch the lips, (C) round the lips and (D) open the lips as much as possible.

1.1.2. Development

Moebius syndrome can present significant challenges in communication, as demonstrated by a study conducted by [2] on the communicative disorders of 22 individuals with the condition. The findings revealed that most individuals exhibited varying degrees of dysarthria. Nonetheless, older children often managed to develop intelligible speech through oral language stimulation and compensatory articulatory strategies. However, children diagnosed with Moebius syndrome frequently experience delays in language development. Research by [3] indicates that these children encounter cognitive and social challenges, including learning difficulties and developmental delays. Despite these obstacles, compelling evidence supports the positive effects of speech therapy interventions on their linguistic skills. This underscores the critical need for targeted interventions that enhance speech production and overall communication abilities in individuals with Moebius syndrome, facilitating better outcomes and improved quality of life.

1.1.3. Articulation and intelligibility

Individuals with Moebius syndrome exhibit distinct articulatory strategies due to their congenital lip paralysis and resulting in incomplete lip closure. These impairments significantly impact the production of bilabial sounds (/m, b, p/), labio-dental sounds (/f, v/), and rounded vowels. Numerous studies have reported frequent misarticulations associated with impaired labial function [4-6]. Interestingly, approximately 20% of individuals with Moebius syndrome have been observed to develop particularly efficient articulatory strategies. These allow them to adapt their speech production despite the challenges posed by their condition.

However, it is noteworthy that to date, there has been no quantitative analysis specifically examining acoustic and articulatory adaptations in individuals with Moebius syndrome who reached intact intelligibility. This research gap highlights the need for further studies to explore the adaptations in speech production within this population.

1.2. Acoustics-articulation

The focus of this study is understanding how different articulatory configurations (here in individuals with congenital lip paralysis) can produce similar acoustic outputs. This phenomenon is crucial for identifying and developing adaptations. For example, alternative articulatory strategies can

allow speakers to develop natural intelligibility even when primary articulators, such as the lips, are compromised. This adaptability, often referred to as motor equivalence [7], underscores the ability of the speech motor system to compensate for changes or perturbations in articulatory movements. Research in this field also highlights the flexibility of the speech motor system, demonstrating how speakers adapt their articulatory movements to achieve intelligible speech despite anatomical or neurological constraints.

Investigating how articulatory constraints affect acoustic outputs is particularly valuable for developing targeted therapies for individuals with speech production challenges. This knowledge is especially relevant for conditions such as Moebius syndrome, where articulatory movements are restricted or impaired. Researchers and clinicians can design more effective interventions to improve communication abilities, already for young individuals with lip paralysis.

1.3. Research objectives

The primary aim of this study is to address the lack of detailed research on speech motor control adaptations in individuals with Moebius syndrome who exhibit complete lip paralysis but achieve intact speech intelligibility. This study represents the first detailed acoustic and articulatory analysis of this population, focusing on how they adapt their speech production to overcome severe articulatory constraints.

2. Methods

2.1. Recordings

2.1.1. Participants

The study included three female French speakers with Moebius syndrome (M1: 41 years, M2: 43 years, M3: 27 years), all of whom demonstrated intact speech intelligibility despite the presence of complete lip paralysis as determined by a speech therapist. To provide a baseline for comparison, the study also included three control speakers without Moebius syndrome. The control group consisted of C1 (28 years, female), C2 (40 years, male), and C3 (20 years, male).

2.1.2. Data collection

Articulatory and acoustic data were collected simultaneously from all six native French speakers. Data collection utilized the Electromagnetic Articulograph AG 501 (Carstens Medizinelektronik GmbH).

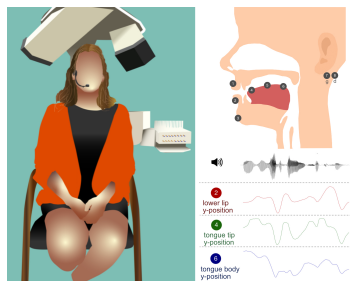


Figure 2: *Experimental setup for recordings with AG501, sagittal view of sensors and extracted trajectories of sensor movements. Figure courtesy by XY.*

Sensors were placed on the chin (CHIN), upper and lower lips (ULIP, LLIP), tongue tip (TTIP), tongue mid (TMID),

and tongue body (TBO), with additional reference sensors positioned behind the left and right ear and the nasion (see Fig. 2). The articulatory signal was recorded at a sampling rate of 1250 Hz. The data were processed using a Butterworth low-pass filter with a cutoff frequency of 25 Hz and filter order 5. Acoustic data were recorded simultaneously at a sampling rate of 44.1 kHz with 16-bit resolution.

2.1.3. Speech material

The speech material analyzed in this study is part of a larger corpus. The current study focuses specifically on word productions to investigate articulatory and acoustic adaptations.

The word production task involved disyllabic target words embedded in carrier phrases, e.g., “Elle a dit *mamie* deux fois.”, Engl. ‘*She said mamie (=granny) two times.*’ These target words were designed to vary systematically in their vowel qualities across both stressed and unstressed syllables. The vowels included in the study were /i, y, e, a, o, u/. These vowels were surrounded by different consonantal contexts /b, m, v/, allowing for an analysis of how articulatory constraints influence speech production and how compensatory strategies are employed. For this analysis, a total of 590 tokens were taken into account, comprising 97 distinct target words.

2.2. Annotations and measurements

Measurements were conducted using a custom script written in Python v3.10.15 [8]. For the acoustic analysis, we used Parselmouth v0.4.3 [9], which interfaces with Praat v6.1.38 [10]. For the articulatory analysis, including the x, y, z-position of sensors, the data extraction routine from ADA (Articulatory Data Analysis; [11]) was applied.

2.2.1. Acoustics measures

For the acoustics analysis, we looked at the spatial and temporal domains. The annotation of the acoustics segments was performed semi-automatically: forced-alignment was conducted using WebMAUS [12] and manually corrected. For the temporal domain, we examined syllable durations relative to the target word duration (unaccented syllable, i.e., C₁V₁ vs. accented syllable, i.e., C₂V₂). For the spatial domain, i.e., formant analysis, we extracted the formants F1, F2 and F3 for the vowels /i, y, e, a, o, u/ at their midpoints (50%) using the Burg algorithm, with the maximum formant adapted to the speakers’ sex (female = 5500Hz, male = 5000Hz) in the unaccented and accented syllables.

2.2.2. Articulatory measures

To shed lights on the modifications in the articulatory domain, we analyzed the sensor’s position on the tongue body (TBO) and tongue tip (TTIP) and the lower lip (LLIP) in 3 dimensions: i.e., the horizontal (x), vertical (y) and the lateral (z) dimensions, for all vowels at their midpoints for the unaccented vs. accented syllables.

2.3. Statistical analyses

Bayesian linear regression models were used for the statistical analysis. Separate models were constructed for each measurement based on vowel position. For formant measures and articulatory measures, we had the following model structure: parameters were entered as response variable and state (control, moebius), vowel (/i, y, e, a, o, u/) and their interaction (state*vowel) as common level effects as well as a

by-speaker intercept as group-level effect. For the parameter of relative syllable duration, a similar structure was used, with state as common-level effect and by-speaker intercept as group-level effect.

The statistical models were executed using Bambi [13], which interfaces with PyMC [14]. We utilized default priors and ran 4 chains, each with 4000 iterations, resulting in a total of 16,000 iterations for the analyses. The articulatory and formant data were z-scored for each speaker prior to the analysis. Results are reported by the mean and the lower and upper bounds of the 95% posterior distribution. Decision-making was guided by the HDI+ROPE decision rule [15], which is based on the relationship between the highest density interval (HDI) of the posterior distribution and a Region of Practical Equivalence (ROPE). Reported are clear differences between control and Moebius speakers when the HDI was entirely outside the ROPE, while no or no clear difference was determined when the HDI included or was inside the ROPE. Reported are also tendencies when the HDI was higher or lower 0. The ROPE was set uniformly to $[-0.1, 0.1]$ for standardized variables following [16].

3. Results

3.1. Acoustics

3.1.1. Syllable duration

First, as expected, the analysis revealed that accented syllables relative to the word duration were longer than unaccented vowels relative to the word duration, in both groups (Moebius: $\text{accented}=0.53 > \text{unaccented}=0.47$ vs. controls: $\text{accented}=0.54 > \text{unaccented}=0.46$). There were no differences in the relative duration of the unaccented and accented syllables when comparing individuals with Moebius syndrome and controls (unaccented syllable: $\beta=0.07 [-1.35, 1.44]$, accented syllable: $\beta=-0.05 [-1.44, 1.36]$).

3.1.2. Formant patterns

For the spatial domain, we looked at formant patterns of the vowels /i, y, e, a, o, u/ in the unaccented and accented domain. Fig. 3 displays the F1-F2 formants comparing individuals with Moebius syndrome (dashed) and control (filled). The analysis revealed that both speaker groups show a similar pattern with rather distinct vowel (both in the unaccented and the accented syllables).

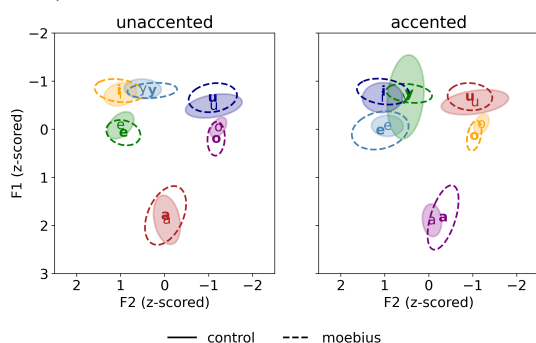


Figure 3: F1-F2 formant plots (z-scored) for unaccented (left) and accented syllables (right) for the vowels /i, y, e, a, o, u/ for individuals with Moebius syndrome (dashed) vs control speakers (filled). Ellipses represent an SD of 1.

Fig. 4 displays the F2-F3 formants comparing individuals with Moebius syndrome (dashed) and control (filled). Again, we see similar patterns when comparing individuals with Moebius syndrome to the control speakers with e.g., F3 being lowered for rounded vowels, for controls, but also for individuals with Moebius syndrome (recall that lip rounding is not possible).

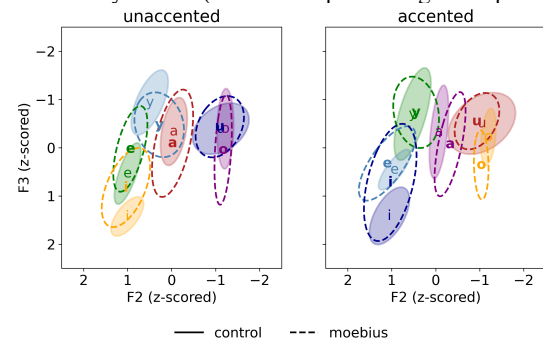


Figure 4: F2-F3 formant plots (z-scored) for unaccented (left) and accented syllables (right) for the vowels /i, y, e, a, o, u/ for individuals with Moebius syndrome (dashed) vs control speakers (filled). Ellipses represent an SD of 1.

The statistical analyses revealed no differences between groups for V1 in the unaccented syllable for F1 ($\beta=-0.09 [-0.23, 0.05]$), for F2 ($\beta=0.03 [-0.15, 0.21]$), and F3 ($\beta=0.24 [-0.0, 0.55]$). Vowel-specific group differences in F3 were found for /e/ ($\beta=-0.51 [-0.83, -0.18]$), /i/ ($\beta=-0.06 [-0.94, -0.28]$) with lower F3 values, and for /o/ ($\beta=0.44 [0.11, 0.79]$), and /y/ ($\beta=0.46 [0.11, 0.79]$) with higher F3 values for individuals with Moebius syndrome.

In V2 (accented syllable), similar non-differences can be reported F1 ($\beta=-0.06 [-0.23, 0.12]$), F2 ($\beta=-0.25 [-0.44, -0.06]$) and also F3 ($\beta=0.26 [-0.08, 0.6]$). For F3, in /i/ with lower values ($\beta=-0.69 [-1.05, -0.3]$) and in /o/ with higher values ($\beta=0.58 [0.26, 0.89]$) for individuals with Moebius syndrome.

3.2. Articulation

In the articulatory domain, we were interested in looking at the positions of the sensors during the vowel production (at 50%) to be comparable to the formant patterns of the individuals with Moebius syndrome and the control speakers. Fig. 5 displays the vertical and horizontal values for each vowel separately for the unaccented and accented syllable for individuals with Moebius syndrome (dashed) and control speakers (filled). By visual inspection, the differences between individuals with Moebius and controls are more pronounced in that there is less overlap between the groups.

We found a tendency for differences between the groups in horizontal position (TBOx) in the V1 (unaccented, $\beta=-0.31 [-0.55, -0.09]$), in that /u/ is more fronted in individuals with Moebius ($\beta=0.63 [0.38, 0.84]$) and /y/ is more back ($\beta=-0.36 [-0.63, -0.11]$) than in controls. Further, there is a difference in the vertical TBO position (TBOy) of individuals with Moebius syndrome in V1 ($\beta=0.74 [0.52, 0.96]$) with a higher TBO for /i/ ($\beta=0.69 [0.46, 0.92]$), and a lower TBO for /o/ ($\beta=-0.73 [-0.97, -0.5]$) and for /u/ ($\beta=-1.12 [-1.35, -0.91]$) when compared to controls. In the lateral dimension (TBOz), there were no differences ($\beta=-0.0 [-0.4, 0.37]$).

In the accented vowel, V2, we did find a difference in the horizontal domain TBOx ($\beta=-0.51 [-0.72, -0.3]$), the vowel /a/ is more back, /u/ is more fronted ($\beta=0.66 [0.43, 0.87]$), /y/ shows

a tendency for more backing ($\beta=-0.24$ [-0.45, -0.02]), and /o/ a tendency for being more fronted in individuals with Moebius syndrome ($\beta=0.22$ [0.01, 0.42]). For the TBOy (vertical) position, we found evidence for a difference in V2 as well, ($\beta=0.75$ [0.52, 0.98]), with a lowered TBO for /o/ ($\beta=-0.51$ [-0.73, -0.27]) and /u/ ($\beta=-0.7$ [-0.94, -0.47]). In the lateral dimension (TBOz) no differences were found ($\beta=0.33$ [-0.07, 0.73]).

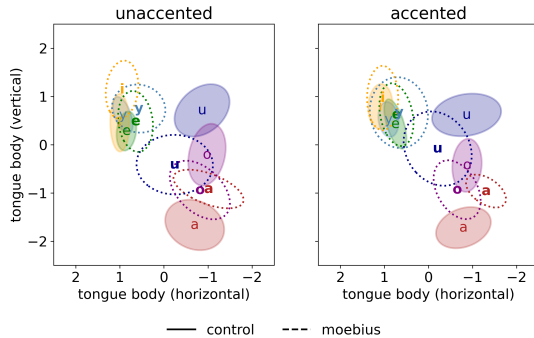


Figure 5: TBOx (vertical) and TBOy (horizontal) positions (z-scored) for unaccented (left) and accented syllables (right) for the vowels /i, y, e, a, o, u/ for individuals with Moebius syndrome (dashed) vs control speakers (filled). Ellipses represent an SD of 1.

To sum up, although we did not find differences in the acoustic formant patterns between the two groups, we observed differences articulatory patterns during the vowel production, particularly in the tongue body vertical and horizontal domain. As has been shown by [17] in that tongue movement during labial fast syllable repetitions are much more pronounced, whereas lip movements much less in individuals with Moebius syndrome than in control speakers, we have also analyzed the position of the tongue tip sensor as well as the lower lip sensor.

For the unaccented V1, we did not find any difference in the LLIPx ($\beta=-0.22$ [-0.48, 0.04]), LLIPy dimension ($\beta=-0.2$ [-0.46, 0.05]), or LLIPz ($\beta=0.04$ [-0.34, 0.43]).

For V2 (accented syllable), we found a difference in the individuals with Moebius syndrome showing lower values in LLIPx ($\beta=-0.34$ [-0.55, -0.15]) and also in LLIPy ($\beta=-0.43$ [-0.65, -0.19]), with no difference in LLIPz ($\beta=-0.14$ [-0.56, 0.24]).

For the unaccented V1, we found a difference in the TTIPx which is more back for individuals with Moebius syndrome ($\beta=-0.57$ [-0.82, -0.33]). We also found a difference in TTIPy with lower tongue tip for individuals with Moebius syndrome ($\beta=-0.48$ [-0.68, -0.28]). Interestingly, there also was a difference in the lateral dimension, TTIPz, being more left oriented during the vowel production ($\beta=1.01$ [0.65, 1.4]).

For the accented V2, we found a difference in the TTIPx which is also more back for individuals with Moebius syndrome ($\beta=-0.84$ [-1.05, -0.62]). Again, we also found a difference in TTIPy with lower tongue tip for individuals with Moebius syndrome ($\beta=-0.72$ [-0.94, -0.49]), and a difference in the lateral dimension, TTIPz, being more left oriented during the vowel production ($\beta=0.68$ [0.27, 1.06]).

4. Discussion and conclusions

This study provides new insights into the acoustic and articulatory adaptations employed by individuals with

congenital lip paralysis (Moebius syndrome) to achieve intelligible speech despite severe articulatory constraints. The key findings are that individuals with Moebius syndrome produce vowel formant patterns that are acoustically very similar to those of control speakers, despite significant differences in their articulatory mechanisms. Specifically, Moebius speakers compensate for their inability to use the lips by employing more pronounced tongue movements, both in the horizontal and vertical dimensions. These adaptations are particularly evident in the production of rounded vowels (with a lowered F3 for rounded vowels), where tongue movements substitute for the lack of lip rounding. Additionally, no significant differences were observed in the temporal domain, as both groups exhibited comparable relative syllable durations for accented and unaccented syllables.

The observed acoustic similarities, and the articulatory uniqueness, align with the concept of motor equivalence, which highlights the flexibility of the speech motor system in achieving similar acoustic outputs through alternative articulatory strategies. This adaptability underscores the remarkable capacity of the human speech motor system to overcome anatomical or neurological constraints. For individuals with Moebius syndrome, the increased reliance on tongue movements compensates for the lip paralysis [17], demonstrating the critical role of the tongue in maintaining intelligibility.

The results of this study have important implications for speech therapy and clinical interventions. The reliance on tongue-based compensatory strategies in individuals with Moebius syndrome highlights the need for therapeutic approaches that focus on enhancing tongue control and flexibility. Speech-language pathologists can use these findings to design targeted interventions that leverage the tongue's adaptability to compensate for lip immobility. For example, therapy could include exercises to improve tongue precision and range of motion, particularly for producing rounded vowels and bilabial sounds. Additionally, understanding the specific articulatory adaptations employed by Moebius speakers can help clinicians develop personalized strategies to optimize speech intelligibility in this population.

While this study provides valuable insights, several limitations should be acknowledged. Future research should also explore potential variability in compensatory strategies across individuals. Additionally, this study focused primarily on vowel production; future research should investigate consonantal features and their associated articulatory adaptations, particularly for bilabial and labiodental sounds, which are most affected by lip paralysis. Detailed analyses of tongue modifications, also in the lateral dimension, could provide deeper insights into the mechanisms underlying motor equivalence.

This study highlights the remarkable adaptability of the speech motor system in individuals with congenital lip paralysis, who achieve intelligible speech despite severe articulatory constraints. The findings emphasize the importance of tongue-based compensatory strategies in overcoming lip immobility and achieving acoustic targets. This research contributes to our understanding of speech production in populations with congenital lip paralysis. These insights have significant implications for clinical practice, offering a foundation for developing targeted interventions to improve communication abilities in individuals with Moebius syndrome and similar conditions.

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